

# Aby Saji Pappachan

**Technical Lead | Backend and Platform Engineering | AI Systems**

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## PROFILE

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Technical Lead with 8 years of experience building and scaling SaaS platforms, backend systems, and data pipelines. Promoted to Technical Lead in 2025 after consistently owning complex projects end-to-end and delivering ahead of schedule. Known across teams as the engineer to bring in when the problem is ambiguous or high-stakes. Strong record of turning engineering work into tangible business outcomes, including ~40% lower infrastructure costs, ~60% faster reporting performance, and improved system reliability. Fully eligible to work in the UK. Currently looking for Backend, AI/LLM, and Platform Engineering roles.

## EXPERIENCE

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### Technical Lead · Josys Digital Technologies

Aug 2022 – Jun 2026

*Previously Senior Software Engineer (Aug 2022 – Mar 2025). Promoted for sustained delivery ownership and architectural leadership across multiple teams. Recognised as a top performer for two consecutive years.*

#### **Data, Reporting and Analytics**

- Designed and delivered a reporting and analytics platform using Airbyte, DuckDB, and Apache Airflow, improving query performance on 50M+-row datasets by ~70% and cutting analytics infrastructure spend by ~45%.
- Took end-to-end ownership of ETL pipeline design and scheduled workflow delivery across 15+ data sources, giving the business reporting at a scale that was not previously cost-effective.
- Optimised analytical queries and led data modelling improvements that reduced turnaround on the most-used business reports from ~25 minutes to under 2 minutes.

#### **AI and LLM Systems**

- Designed and shipped AI-powered internal platforms using LLMs for context-aware data retrieval, cutting the time engineers spent manually querying and correlating data by ~60% (around 8 hours/week saved).
- Introduced Graph Database-backed solutions to model complex entity relationships across 20+ entity types, enabling semantic querying the relational setup could not support.
- Built MCP (Model Context Protocol) clients and servers to standardise how AI agents communicate with backend services, cutting integration time for new services from ~2 weeks to ~2 days.
- Worked LLM workflows into existing products to automate decision-making steps, removing around 10 hours/week of manual intervention from the team.

#### **Backend and Platform Engineering**

- Championed NestJS as the company-wide backend standard across 12+ services, reducing new-engineer onboarding from ~6 weeks to ~3 weeks and improving long-term maintainability.
- Designed and shipped a Notification Platform using REST APIs, AWS SQS, Email, and Slack integrations, consolidating 8 fragmented notification flows into one reliable system handling ~50,000 notifications/day.
- Built a Performance Testing Framework that simulated large-scale SaaS environments, cutting testing infrastructure costs by ~35% and catching reliability issues before they reached production.
- Delivered a multi-database GraphQL platform with automatic schema generation and dynamic filter conversion, removing ~70% of repetitive boilerplate and speeding up feature delivery.
- Built a cross-browser extension backend supporting both Chromium and Firefox with dynamic configuration, unblocking a product feature that had stalled on browser compatibility constraints.
- Created proxy servers that decoupled frontend and backend dependencies, speeding up UI team delivery cycles by ~30% across 5 projects.

- Regularly brought in for technical research, architecture reviews, and proof-of-concept work outside the scope of individual projects.

#### **Member of Technical Staff (MTS-1) · InstaSafe**

May 2019 – Aug 2022

- Led a team of 3 engineers delivering 6 security-focused product enhancements, with full ownership from design through to production.
- Designed and built a Zero Trust Network Access (ZTNA) solution from scratch, which became a new product capability in the company's enterprise offering and contributed to onboarding 10+ enterprise customers.
- Shipped a cross-platform desktop application using ElectronJS for secure access to private resources, adopted by ~5,000 users.
- Built a gateway server enabling agentless access to private web applications, reducing installation friction and cutting enterprise customer onboarding time by ~40%.
- Implemented secure CRUD APIs and automated workflows using Node.js and Spring Boot.

#### **Software Engineer · GrowthEnabler**

Aug 2018 – May 2019

- Worked as a full-stack engineer across frontend and backend, building an SRM workflow platform and a Solution Source system to manage startup and corporate collaborations.

#### **Developer · Haappy**

Jan 2018 – Jul 2018

- Developed multiple WebTV websites and a CRM system for task and workflow tracking.

### **TECHNICAL SKILLS**

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**Languages and Frameworks:** Node.js, TypeScript, NestJS, Python (FastAPI), Golang, GraphQL, REST APIs

**Cloud and Infrastructure:** AWS (ECS, SQS, S3), Docker, Cloud-native architectures

**Data and Analytics:** Airbyte, DuckDB, Apache Airflow, ETL pipelines, Analytical data modelling

**AI and Intelligent Systems:** LLMs, LangChain, Graph Databases, MCP (Model Context Protocol), Semantic search

**Databases:** MySQL, PostgreSQL, MongoDB, Graph Databases

**Engineering Practices:** Distributed systems, System design, Performance testing, Backend architecture

**Frontend (Secondary):** React, Angular

### **EDUCATION**

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**Bachelor of Engineering, Computer Science** 2013 – 2017

East Point College of Engineering and Technology, Bengaluru, India